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Practical :-1

Aim:- Study of different network devices in detail.

1. Network Interface Card (NIC)

- **Definition:**
A Network Interface Card is a hardware component that connects a computer to a network.
- Functions:
 - Converts data from the computer into signals for transmission
 - Provides a unique MAC address
 - Controls data flow between computer and network
- Types:
 - Wired NIC (Ethernet)
 - Wireless NIC (Wi-Fi)
- Advantages:
 - Essential for network connectivity
 - High-speed data transmission

2. Repeater

- **Definition:**
A repeater is a device that regenerates and amplifies signals to extend the distance of a network.
- Functions:
 - Receives weak signals
 - Amplifies and retransmits them
- OSI Layer: Physical Layer
- Advantages:
 - Extends network coverage
 - Simple and low cost
- Disadvantages:
 - Cannot filter traffic
 - Increases network congestion

3. Hub

- **Definition:**
A hub is a basic networking device that connects multiple computers in a network.
- Functions:
 - Broadcasts data to all connected devices
- Types:
 - Passive Hub
 - Active Hub
 - Intelligent Hub
- OSI Layer: Physical Layer
- Disadvantages:
 - High collision rate
 - Low security

- Inefficient

4. Bridge

- **Definition:**
A bridge connects two or more LAN segments and filters traffic based on MAC addresses.
- Functions:
- Reduces network traffic
- Filters frames
- OSI Layer: Data Link Layer
- Advantages:
- Improves performance
- Reduces collisions
- Disadvantages:
- Slower than switches
- Limited scalability

5. Switch

- **Definition:**
A switch is an advanced version of a bridge that connects multiple devices and forwards data only to the destination device.
- Functions:
- Uses MAC address table
- Supports full-duplex communication
- OSI Layer: Data Link Layer (Layer 2)
- Advantages:
- High speed
- Low collisions
- Better security than hubs
- Types:
- Managed Switch
- Unmanaged Switch
- Layer 3 Switch

6. Router

- **Definition:**
A router connects different networks and routes data packets using IP addresses.
- Functions:
- Determines best path for data
- Connects LAN to WAN (Internet)
- OSI Layer: Network Layer
- Advantages:
- Reduces broadcast traffic
- Provides security features (firewall, NAT)

- Disadvantages:
- More expensive than switches
- Configuration complexity

7. Gateway

- **Definition:**
A gateway connects networks using different protocols and performs protocol conversion.
- Functions:
- Translates data formats
- Acts as entry/exit point of a network
- OSI Layer: All layers
- Advantages:
- Enables communication between different systems
- Example:
Email gateway, VoIP gateway

8. Modem

- **Definition:**
A modem converts digital data into analog signals and vice versa.
- Functions:
- Modulation
- Demodulation
- Types:
- Dial-up Modem
- DSL Modem
- Cable Modem
- Fiber Modem
- Advantages:
- Internet connectivity over telephone/cable lines

9. Wireless Access Point (WAP)

- **Definition:**
A WAP allows wireless devices to connect to a wired network.
- Functions:
- Provides Wi-Fi connectivity
- Manages wireless clients
- Advantages:
- Mobility
- Easy installation

10. Firewall

- **Definition:**
A firewall is a security device that monitors and controls network traffic.

- Functions:
- Blocks unauthorized access
- Allows authorized traffic
- Types:
- Hardware Firewall
- Software Firewall
- Advantages:
- Improves network security
- Protects against attacks

11. Proxy Server

- **Definition:**
A proxy server acts as an intermediary between users and the internet.
- Functions:
- Filters requests
- Caches web pages
- Provides anonymity
- Advantages:
- Faster access
- Better security

12. Network Devices Comparison Table

Device	OSI Layer	Main Function
NIC	Physical/Data Link	Network connection
Repeater	Physical	Signal regeneration
Hub	Physical	Broadcast data
Bridge	Data Link	Traffic filtering
Switch	Data Link	Frame forwarding
Router	Network	Packet routing
Gateway	All layers	Protocol conversion
Modem	Physical	Signal conversion
Firewall	Network/Application	Security
WAP	Data Link	Wireless access

Practical :-2

Aim:- Study of different types of network cables and practically implement the cross-wired cable and straight through cable using clamping tool.

1. Study: Types of Network Cables**A. Straight-Through Cable**

- Purpose: Connects different devices.
 - Example: Computer → Switch, Router → PC
- Wiring: Same pin configuration on both ends.
- Standard:
 - TIA/EIA 568B or 568A (most use 568B)
- Color Code Example (568B):
- Pin: 1 2 3 4 5 6 7 8
- Wire: Orange-white, Orange, Green-white, Blue, Blue-white, Green, Brown-white, Brown.

B. Crossover Cable

- Purpose: Connects similar devices.
 - Example: PC → PC, Switch → Switch
- Wiring: Transmit and receive pairs are swapped.
 - One end: 568A, other end: 568B
- Color Code Example:
 - End 1 (568A): Green-white, Green, Orange-white, Blue, Blue-white, Orange, Brown-white, Brown
 - End 2 (568B): Orange-white, Orange, Green-white, Blue, Blue-white, Green, Brown-white, Brown

C. Tools Needed

1. RJ45 connectors
2. Clamping/Crimping tool
3. Cable stripper / cutter
4. Cable tester (optional but recommended)

2. Practical Implementation

Step 1: Prepare the Cable

1. Cut the cable to the required length.
2. Strip about 2–3 cm of the outer sheath using a cable stripper.
3. Untwist the pairs and straighten the wires.

Step 2: Arrange the Wires

- For Straight-Through: Both ends the same (e.g., 568B → 568B)
- For Crossover: One end 568A, other end 568B
- Tip: Keep the wires flat and in correct order.

Step 3: Trim the Wires

- Trim wires to approx. 1.5 cm so they fit evenly into the RJ45 connector.

Step 4: Insert Wires into RJ45 Connector

- Ensure the metal contacts face up, wires fully reach the end of the connector.

Step 5: Crimp the Connector

1. Place connector in the crimping tool.
2. Squeeze the handle firmly to fix the wires.
3. Repeat for the other end.

Step 6: Test the Cable

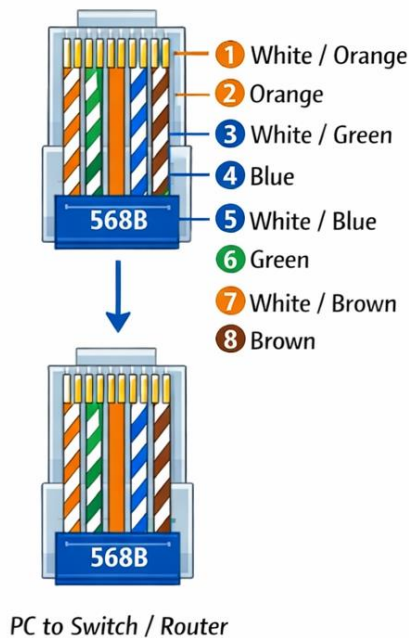
- Use a cable tester to ensure correct wiring.
 - Straight-through: Pins 1→1, 2→2, etc.
 - Crossover: Pins 1→3, 2→6, 3→1, 6→2 (others same)

4. Summary Table

Cable Type	Use Case	Wiring Standard
Straight-Through	PC → Switch, Router → PC	568B → 568B
Crossover	PC → PC, Switch → Switch	568A → 568B

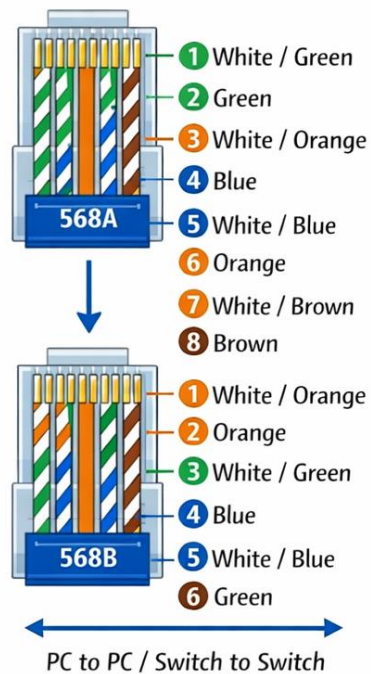
STRAIGHT-THROUGH CABLE

Same on Both Ends (568B - 568B)



CROSSOVER CABLE

Different on Each End (568A - 568B)



1 White / Orange 2 Orange 3 White / Green 4 Blue 5 White / Blue 6 Green 7 White / Brown 8 Brown

Practical :-3

Aim:- Study of basic network command and Network configuration commands.

1. Basic Network Commands

- These are commands mainly used to check, diagnose, and troubleshoot network connections.

a) ping

- Purpose: Tests connectivity between your device and another device (host).
- Example:
- ping google.com
- Notes:
 - Returns response time.
 - If it fails, there may be network issues.

b) ipconfig (Windows) / ifconfig (Linux)

- Purpose: Shows network configuration of your system.
- Example (Windows):
- ipconfig
- Example (Linux):
- ifconfig
- Notes:
 - Shows IP address, subnet mask, gateway, and DNS.
 - ipconfig /all gives full details.

c) tracert (Windows) / traceroute (Linux)

- Purpose: Shows the path packets take to reach a host.
- Example:
- tracert google.com
- Notes: Helps identify where connection is failing.

d) nslookup

- Purpose: Queries DNS to find IP addresses for a domain.
- Example:
- nslookup google.com
- Notes: Useful for checking if DNS resolution works.

e) netstat

- Purpose: Shows active network connections and listening ports.
- Example:
- netstat -an
- Notes: Useful for network troubleshooting or detecting suspicious connections.

f) arp

- Purpose: Shows MAC addresses associated with IPs in your local network.
- Example:
- arp -a

- Notes: Helps identify devices on the same LAN.

g) hostname

- Purpose: Displays your device's network name.
- Example:
- Hostname

h) route

- Purpose: Displays or modifies routing tables.
- Example:
- route print (Windows)
- route -n (Linux)

2. Network Configuration Commands

- These commands are used to set or change network settings such as IP, subnet mask, gateway, and DNS.

a) Set IP address manually

Windows

```
netsh interface ip set address "Ethernet" static 192.168.1.10 255.255.255.0 192.168.1.1
```

- "Ethernet" = network interface name
- 192.168.1.10 = IP address
- 255.255.255.0 = Subnet mask
- 192.168.1.1 = Default gateway

Linux

```
sudo ifconfig eth0 192.168.1.10 netmask 255.255.255.0
```

```
sudo route add default gw 192.168.1.1
```

- eth0 = network interface name

b) Set DNS servers

Windows

```
netsh interface ip set dns "Ethernet" static 8.8.8.8
```

Linux

- Edit /etc/resolv.conf:

```
nameserver 8.8.8.8
```

```
nameserver 8.8.4.4
```

c) Enable/Disable network interface

Windows

```
netsh interface set interface "Ethernet" admin=disable
```

```
netsh interface set interface "Ethernet" admin=enable
```

Linux

```
sudo ifconfig eth0 down
```

```
sudo ifconfig eth0 up
```

d) DHCP (Dynamic IP)

```
netsh interface ip set address "Ethernet" dhcp
```

Linux

```
sudo dhclient eth0
```

Summary Table

Command	Purpose	Example
ping	Checks connectivity between two devices	ping google.com
ipconfig (Windows) / ifconfig (Linux)	Displays IP configuration details	ipconfig /all
tracert (Windows) / traceroute (Linux)	Shows the route taken by packets to reach a destination	tracert google.com
nslookup	Checks DNS resolution of a domain name	nslookup google.com
netstat	Displays active network connections and ports	netstat -an
arp	Shows MAC address to IP address mapping	arp -a
hostname	Displays the name of the device	hostname
route	Displays or modifies the routing table	route print
netsh (Windows) / ifconfig (Linux)	Configures network interfaces, IP, DNS	netsh interface ip set address "Ethernet" static ...
DHCP commands	Obtains IP address dynamically from DHCP server	dhclient eth0

Practical :-4

Aim:- Implement different LAN topologies using Network Simulator.

1. Bus Topology

- Description: All devices share a single communication line (backbone). Data sent by one device is received by all others.
- Implementation Steps in Packet Tracer:
 1. Place multiple PCs on the workspace.
 2. Connect all PCs using a single coaxial cable or switch (simulated bus with a hub if switch not allowed).
 3. Assign IP addresses to all devices.
 4. Test connectivity using Ping.

2. Star Topology

- Description: All devices connect to a central hub or switch. Very common in modern LANs.
- Implementation Steps:
 1. Place a switch in the center.
 2. Connect each PC to the switch using cables (usually copper straight-through).
 3. Assign IP addresses.
 4. Test connectivity (Ping from PC1 to PC2, etc.).

3. Ring Topology

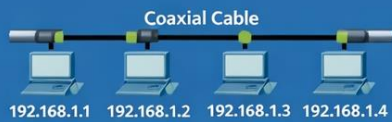
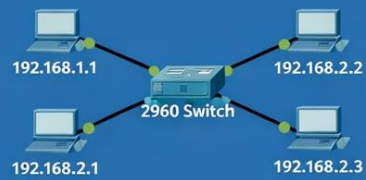
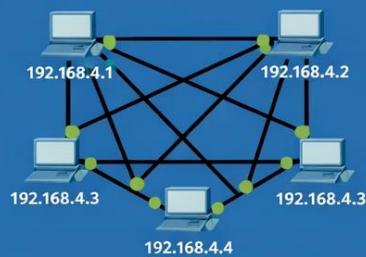
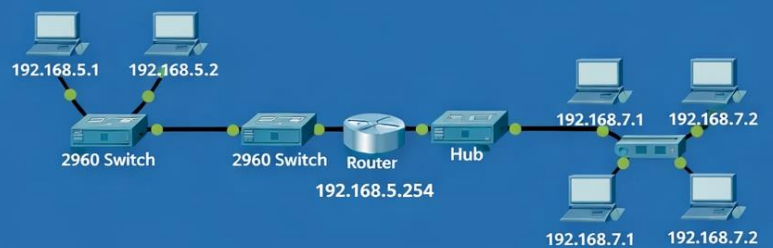
- Description: Each device connects to exactly two other devices, forming a ring. Data travels in one direction.
- Implementation Steps:
 1. Place PCs in a circular layout.
 2. Connect each PC to the next using cables.
 3. Connect the last PC back to the first to complete the ring.
 4. Assign IPs and test connectivity.

4. Mesh Topology

- Description: Every device is connected to every other device. Highly fault-tolerant.
- Implementation Steps:
 1. Place multiple PCs.
 2. Connect every PC to every other PC.
 3. Assign IP addresses.
 4. Test connectivity across all devices.

5. Hybrid Topology

- Description: Combination of two or more topologies. E.g., star-bus or star-ring.
- Implementation Steps:
 1. Decide which topologies to combine (e.g., 3 stars connected in a bus).
 2. Implement each small topology separately.
 3. Interconnect them according to your hybrid design.
 4. Assign IP addresses and test connectivity.

LAN TOPOLOGY SIMULATIONS IN PACKET TRACER**Bus Topology****Star Topology****Ring Topology****Mesh Topology****Hybrid Topology**

Practical :-5

Aim:- Implement the concept of VLAN using Network Simulator.

Step 1: Setup the Network Topology

1. Open Packet Tracer.
2. Drag the following devices:
 - 1 or 2 Switches (e.g., 2960).
 - 4 PCs (to assign to different VLANs).
 - Optionally, 1 Router (if you want inter-VLAN communication later).

• Topology Example:

```
PC1 ----|
PC2 ----|---- Switch
PC3 ----|
PC4 ----|
```

Step 2: Create VLANs on the Switch

1. Click on the Switch → Go to CLI tab.
2. Enter privileged EXEC mode:

```
Switch> enable
```

```
Switch# configure terminal
```

3. Create VLANs (e.g., VLAN 10 for PC1 & PC2, VLAN 20 for PC3 & PC4):

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Sales
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name Marketing
```

```
Switch(config-vlan)# exit
```

Step 3: Assign Ports to VLANs

• Assume:

- PC1 → FastEthernet 0/1
- PC2 → FastEthernet 0/2
- PC3 → FastEthernet 0/3
- PC4 → FastEthernet 0/4

• Assign VLANs to these ports:

```
Switch(config)# interface fa0/1
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```

```
Switch(config-if)# exit
```

```
Switch(config)# interface fa0/2
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```

```
Switch(config-if)# exit
```

```
Switch(config)# interface fa0/3
```

```
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
```

```
Switch(config)# interface fa0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
```

Step 4: Assign IP Addresses to PCs

- Assign IP addresses on each PC to match their VLAN subnet:

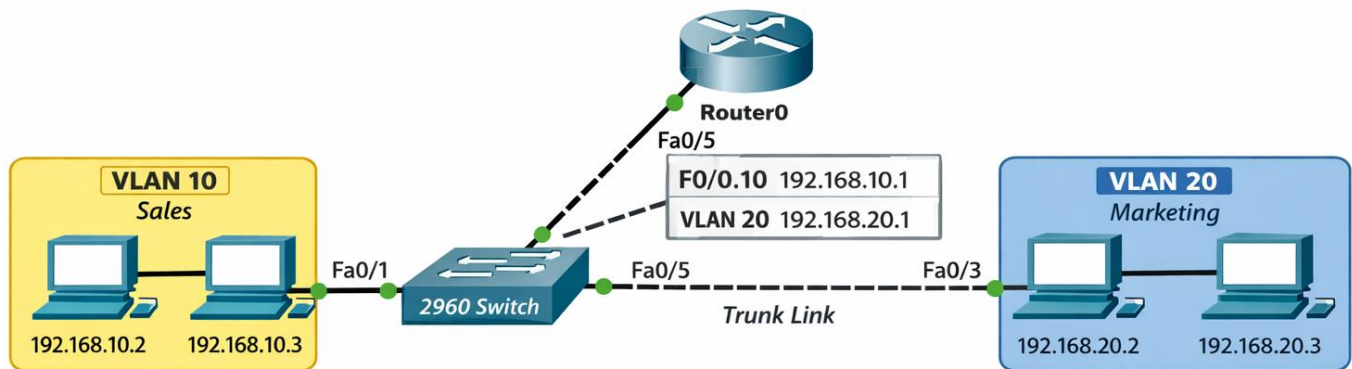
Device	VLAN	IP Address	Subnet Mask
PC1	10	192.168.10.2	255.255.255.0
PC2	10	192.168.10.3	255.255.255.0
PC3	20	192.168.20.2	255.255.255.0
PC4	20	192.168.20.3	255.255.255.0

- Click on each PC → Desktop → IP Configuration → Enter IP and Subnet Mask.

Step 5: Test VLAN Configuration

1. Ping within the same VLAN:
 - PC1 → PC2 → Should succeed
 - PC3 → PC4 → Should succeed
2. Ping across VLANs:
 - PC1 → PC3 → Should fail (different VLANs, no router yet)

VLAN Configuration & Inter-VLAN Routing



VLAN 10 Network: 192.168.10.0 / 24

→ Ping: PC1 → PC2

VLAN 20 Network: 192.168.20.0 / 24

→ Ping: PC1 ✗ PC3

Practical :-6

Aim:- Implement the concept of static routing.

- To implement static routing and enable communication between different networks using manually configured routes.

Theory (Static Routing)

- Static routing is a routing technique in which routes are manually configured by the network administrator.
The router does not automatically update routes if the network topology changes.
- Syntax (Cisco Router):

ip route <destination-network> <subnet-mask> <next-hop-ip / exit-interface>

Network Topology

PC1 ---- R1 ---- R2 ---- PC2

IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask
PC1	NIC	192.168.1.2	255.255.255.0
R1	Fa0/0	192.168.1.1	255.255.255.0
R1	Fa0/1	10.0.0.1	255.255.255.252
R2	Fa0/0	10.0.0.2	255.255.255.252
R2	Fa0/1	192.168.2.1	255.255.255.0
PC2	NIC	192.168.2.2	255.255.255.0

Implementation Steps

Step 1: Configure PCs

- PC1
 - IP: 192.168.1.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
- PC2
 - IP: 192.168.2.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.2.1

Step 2: Configure Router R1

```
enable
configure terminal
interface fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

```
interface fa0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
```

exi

Step 3: Configure Router R2

enable

configure terminal

interface fa0/0

ip address 10.0.0.2 255.255.255.252

no shutdown

interface fa0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

Step 4: Configure Static Routes

- On Router R1

ip route 192.168.2.0 255.255.255.0 10.0.0.2

- On Router R2

ip route 192.168.1.0 255.255.255.0 10.0.0.1

Step 5: Verification

- Check routing table

show ip route

- Test connectivity

- From PC1:

ping 192.168.2.2

- From PC2:

ping 192.168.1.2

- Successful replies confirm static routing is working.

Advantages

- Simple to configure
- Secure (no routing updates shared)
- Low CPU and bandwidth usage

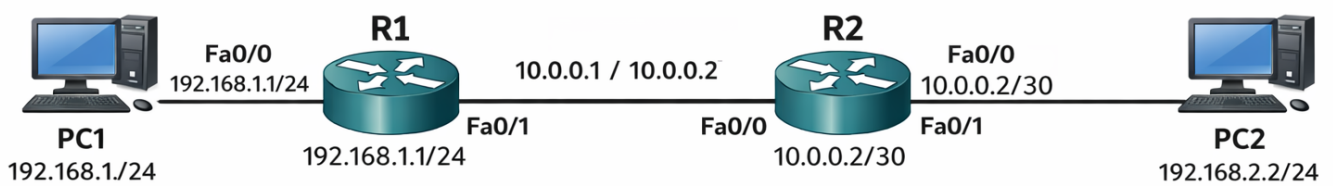
Disadvantages

- Not scalable
- Manual update required if topology changes
- Not suitable for large networks

Result

- Thus, static routing was successfully implemented, enabling communication between two different networks using manually configured routes.

Static Routing



On Router R1

```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

On Router R2

```
ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Practical :-7**Aim:- Implement the concept of dynamic routing.(RIP, OSPF, BGP)****1. Concept of Dynamic Routing**

- Dynamic routing is a routing technique in which routers automatically learn and update routes using routing protocols.
Routers exchange routing information and choose the best path based on metrics like hop count, cost, bandwidth, etc.

Advantages

- Automatically updates routes
- Suitable for large networks
- Handles link failures efficiently

Disadvantages

- Uses more CPU and bandwidth
- More complex than static routing

2. Types of Dynamic Routing Protocols

Protocol	Type	Metric	Standard
RIP	Distance Vector	Hop Count	IGP
OSPF	Link State	Cost	IGP
BGP	Path Vector	AS Path	EGP

3. RIP (Routing Information Protocol)**Features**

- Distance Vector protocol
- Uses Hop Count as metric
- Maximum hop count = 15
- Periodic updates every 30 seconds
- Suitable for small networks

RIP Implementation (Cisco Commands)

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# router rip
```

```
Router(config-router)# version 2
```

```
Router(config-router)# network 192.168.1.0
```

```
Router(config-router)# network 192.168.2.0
```

```
Router(config-router)# no auto-summary
```

```
Router(config-router)# exit
```

Verification

```
Router# show ip route
```

```
Router# show ip protocols
```

4. OSPF (Open Shortest Path First)**Features**

- Link State protocol

- Uses Cost as metric
- Faster convergence
- Supports large networks
- Uses Areas (Area 0 = backbone)

OSPF Implementation (Cisco Commands)

Router> enable

Router# configure terminal

Router(config)# router ospf 1

Router(config-router)# network 192.168.1.0 0.0.0.255 area 0

Router(config-router)# network 192.168.2.0 0.0.0.255 area 0

Router(config-router)# exit

Verification

Router# show ip ospf neighbor

Router# show ip route ospf

5. BGP (Border Gateway Protocol)

Features

- Path Vector protocol
- Used between Autonomous Systems
- Internet backbone protocol
- Uses AS Path as metric
- Very scalable and reliable

BGP Implementation (Cisco Commands)

Router> enable

Router# configure terminal

Router(config)# router bgp 65001

Router(config-router)# neighbor 192.168.10.2 remote-as 65002

Router(config-router)# network 10.0.0.0 mask 255.0.0.0

Router(config-router)# exit

Verification

Router# show ip bgp

Router# show ip bgp summary

6. Comparison Diagram (Textual)

RIP → Small networks → Hop Count → Slow

OSPF → Medium/Large → Cost → Fast

BGP → Internet → AS Path → Very Large

7. Conclusion

- RIP is simple but limited
- OSPF is efficient and widely used in enterprises
- BGP is used for routing between ISPs and large organizations
- Dynamic routing makes networks adaptive, scalable, and fault-tolerant.

Practical :-8

Aim:- Packet capture and header analysis by wire-shark (TCP,UDP,IP)

- To capture network packets using Wireshark and analyze the IP, TCP, and UDP protocol headers.

Software Required

- Wireshark (latest version)
- Operating System: Windows / Linux
- Active Network Connection (LAN / Wi-Fi)

Theory

What is Wireshark?

- Wireshark is a network protocol analyzer used to capture, inspect, and analyze packets traveling through a network in real time.

Protocol Overview

1. IP (Internet Protocol)

- Provides logical addressing
- Responsible for packet routing
- Works at Network Layer

2. TCP (Transmission Control Protocol)

- Connection-oriented
- Reliable data delivery
- Uses 3-way handshake

3. UDP (User Datagram Protocol)

- Connectionless
- Faster but unreliable
- Used in streaming, DNS, VoIP

Procedure

Step 1: Start Packet Capture

1. Open Wireshark
2. Select network interface (Wi-Fi / Ethernet)
3. Click Start Capture

Step 2: Generate Traffic

- Open a website (for TCP)
- Use ping command (for IP)
- Open YouTube or DNS query (for UDP)

Step 3: Apply Filters

- Use these filters in Wireshark:

Protocol	Filter
IP	ip
TCP	tcp
UDP	udp

Header Analysis

IP Header Fields

Field	Description
Version	Indicates IP version (IPv4 or IPv6)
Header Length	Specifies the size of the IP header
Source IP	IP address of the sending device
Destination IP	IP address of the receiving device
TTL (Time To Live)	Limits the lifetime of a packet to prevent looping
Protocol	Indicates the upper-layer protocol (TCP, UDP, etc.)

TCP Header Fields

Field	Description
Source Port	Port number of the sending application
Destination Port	Port number of the receiving application
Sequence Number	Specifies the order of packets in the data stream
Acknowledgment Number	Confirms receipt of data and indicates the next expected sequence number
Flags	Control bits such as SYN, ACK, FIN used for connection management
Window Size	Specifies the amount of data that can be received before acknowledgment (flow control)

UDP Header Fields

Field	Description
Source Port	Port number of the sending application
Destination Port	Port number of the receiving application
Length	Total length of UDP header and data
Checksum	Used for error detection in UDP header and data

Sample Observations

TCP Packet

- Protocol: TCP
- Source Port: 443
- Destination Port: Random
- Flags: SYN, ACK
- Header Length: 20 bytes

UDP Packet

- Protocol: UDP
- Source Port: 53 (DNS)
- Destination Port: Random
- Length: 64 bytes

Result

- Network packets were successfully captured using Wireshark and IP, TCP, and UDP headers were analyzed in detail.